



TEKNOLOGI INFORMASI SISTEM ADMINISTRASI JARINGAN

IT NETWORK SYSTEMS ADMINISTRATION

Actual Task

**LOMBA KOMPETENSI SISWA SEKOLAH MENENGAH KEJURUAN
TINGKAT DKI Jakarta Wilayah Jakarta Timur Tahun 2024**

Module Client Server

Basic Configuration

- Configure username and password login OS according to appendix credential
- Configure all server with hostname and IP Address according to appendix IP Address configuration table
- Configure all linux server to allow non-root login via SSH
- Configure all windows server to be pingable

Supporting Services

DHCP

- NF will serve DHCP Server. You can freely use any tools to provide the service
- Enable DHCP Relay in EDGE to forward DHCP requests to NF
- Configure DHCP to provide addresses to the DMZ zone. Use static mapping for all servers according to appendix table
- Configure DHCP to provide addresses to the Internal zone. Use available unused IP ranges in the subnet, refer to appendix table

Firewall & Reverse Proxy

- Configure NF using iptables LOG module, log following traffics:
 - Log outgoing traffic coming from this server's DHCP port to Internal zone
 - Log incoming HTTPS traffic
 - Put all log in single file: /var/log/firewall.log
- Configure NF using haproxy as reverse proxy engine for incoming HTTPS traffic to MON servers
 - Get the required certificate to enable HTTPS
 - Make sure icinga monitoring web UI is accessible via this service
- Create simple firewall rule in EDGE to block these traffic:
 - From NF to SERVICE using HTTP and HTTPS port

Routing

- Enable routing in NF & EDGE
- Route traffic from Internal to NF, and vice-versa via EDGE
 - Do not route traffic from ops
- Route traffic from DMZ to NF, and vice-versa via EDGE
 - Do not route traffic from ops

Monitoring

- Configure centralized monitoring using icinga2 hosted in MON
 - Enable web interface, make it accessible using port 80
 - Make sure it's accessible via mon.itnsa.id
- Configure monitoring for services and servers
 - Add mail and webmail services to be monitored in icinga2
 - Add SSH services in all linux servers in both Internal and DMZ to be monitored in icinga2
- Configure monitoring for failed SSH Logins in all linux servers
 - Configure mail notification to ops@itnsa.id whenever someone fails to login using SSH

Email

- Configure MAIL as centralized mail server using any application that supports SMTP and IMAP using negotiable TLS
 - Use the domain itnsa.id so mail can be send directly to @itnsa.id mail address
 - Configure SMTP to listen in port 25
 - Enable negotiable TLS using certificate from Corporate CA
 - Configure IMAP to listen in port 143
 - Enable negotiable TLS using certificate from Corporate CA
- Enable web-based email using roundcube.
 - Enable https access using certificate from Corporate CA
 - Make it accessible with the domain webmail.itnsa.id
- Configure Mail Users according to table in the appendix
- Configure Mail Groups notification@itnsa.id with following members:
 - ops@itnsa.id
 - dev@itnsa.id

Certificate Authority

- Configure MAIL as Root CA
 - Use Common Name: LKS2024-Root
 - Approve Intermediate CA Requests for MON
 - Save those two Intermediate CA certificate files without the key in directory /backup in MAIL server
- Configure MON as Intermediate CA Issuer
 - Use Common Name: LKS2024-Intermediate
- In any of the Intermediate CA, issue the certificates required for other services
 - For record, place all generated certificates in /backup/certs in MAIL server

Main Services

VPN and Virtual Users

- Configure LDAP in MAIL to provide users available for VPN Authentication
 - Configure using domain dc=itnsa,dc=id
 - Create user 'vpn' with password 'Skills40' for VPN testing
- Configure openvpn in MON to provide remote access VPN to remote clients.
 - Allow any client to connect using username and password authentication via LDAP
 - Distribute client configuration file to connect in /etc/openvpn/client.ovpn
 - Configure remote clients to use following IP:
 - Start: 10.20.22.10
 - End: 10.22.22.50
 - Subnet: 255.255.255.0
 - Gateway: 10.22.22.1
- You can use NF to test the VPN Client connection, but please don't keep the connection running

Main Website

- Configure SERVICE to hosts website:
 - managers.itnsa.id at C:\web\managers
 - dev.itnsa.id at C:\web\dev
- Enable HTTPS for managers.itnsa.id and dev.itnsa.id
 - Use Corporate CA that points to wildcard domain of *.itnsa.id
- Refer to the appendix for website content.
- Make sure the DNS record is also created at Main DNS.

Main DNS

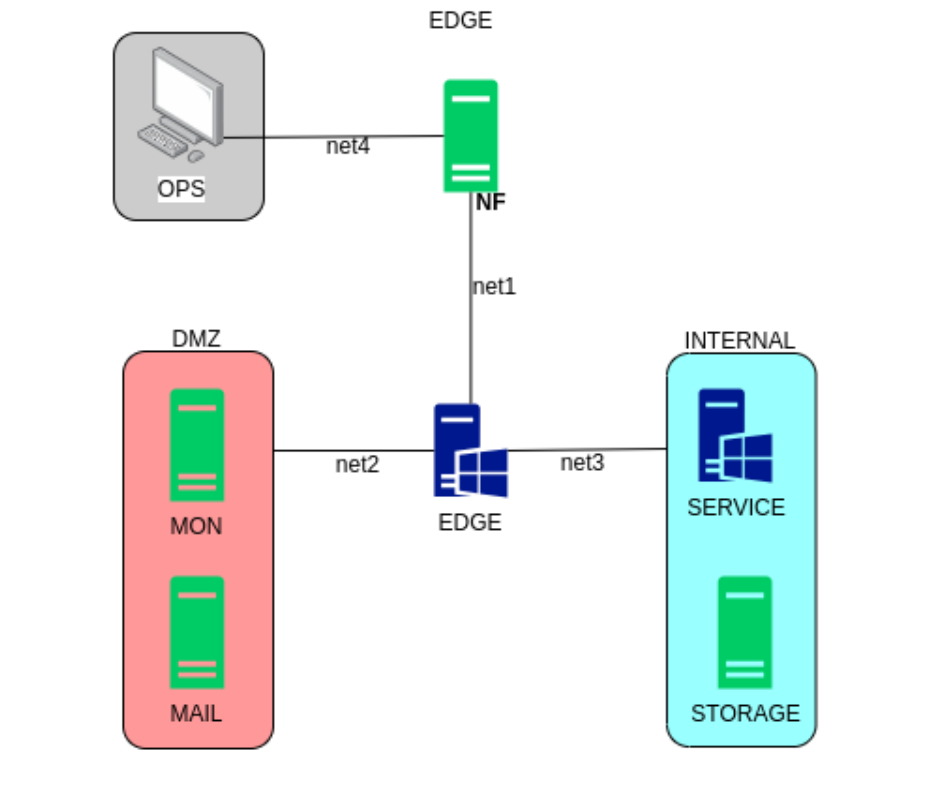
- Configure SERVICE to serve DNS for all itnsa.id domains
 - Add all servers' hostname to be accessible via {hostname}.itnsa.id
 - The subdomain points to all available IP addresses of the servers according to the appendix
 - Refer to other tasks for required records, including but not limited to:
 - Email
 - All Web Domains
 - Monitoring
 - Set NS record for the domain to SERVICE

Shared Folder

- Configure STORAGE to host a CIFS shared folder that can be mounted at Windows Server.
 - Use samba or any other similar application.
 - Use Directory: /share
 - Allow all anonymous to read and write to the directory.
- Create a backup job using the windows server backup feature.
 - Backup C:\web in STORAGE to the shared folder daily at any hour.
 - Make sure the backup is successfully executed at least once.

Appendix

Topology



Credentials

Debian Server : • username : root • password : Skills40 • Username : student • Password : Skills40	Debian Desktop : • username : student • password : Skills40 Windows Server • username : Administrator • password : Skills40
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IP Address Configuration Table

Hostname	Operating System	IP Address
nf	Linux Debian 10 Server	10.199.99.254/28
		202.29.195.1/30
edge	Windows Server 2019 Desktop	10.199.99.253/28
		10.99.99.253/24
		172.16.34.125/24
mon	Linux Debian 10 Server	10.99.99.21/24
mail	Linux Debian 10 Server	10.99.99.25/24
service	Windows Server 2019 Desktop	172.16.34.11/24
storage	Linux Debian 10 Server	172.16.34.12/24
ops	Linux Debian 10 Desktop	202.29.195.2/30

Mail User

Email	Password	Group
ops@itnsa.id	Skills40	notification@itnsa.id
dev@itnsa.id	Skills40	notification@itnsa.id
admin@itnsa.id	Skills40	notification@itnsa.id

Website content

managers.itnsa.id

```
<h1> managers.itnsa.id </h1>
This website is managed by admin@itnsa.id
```

dev.intsa.id

```
<h1> dev.itnsa.id </h1>
This website is managed by dev@itnsa.id
```